Advanced Computer-Aided VLSI System Design
Homework 4 Report

Due Tuesday, May 26, 13:59

Student ID:

Student Name:

APR Results

1. Fill in the blanks below.

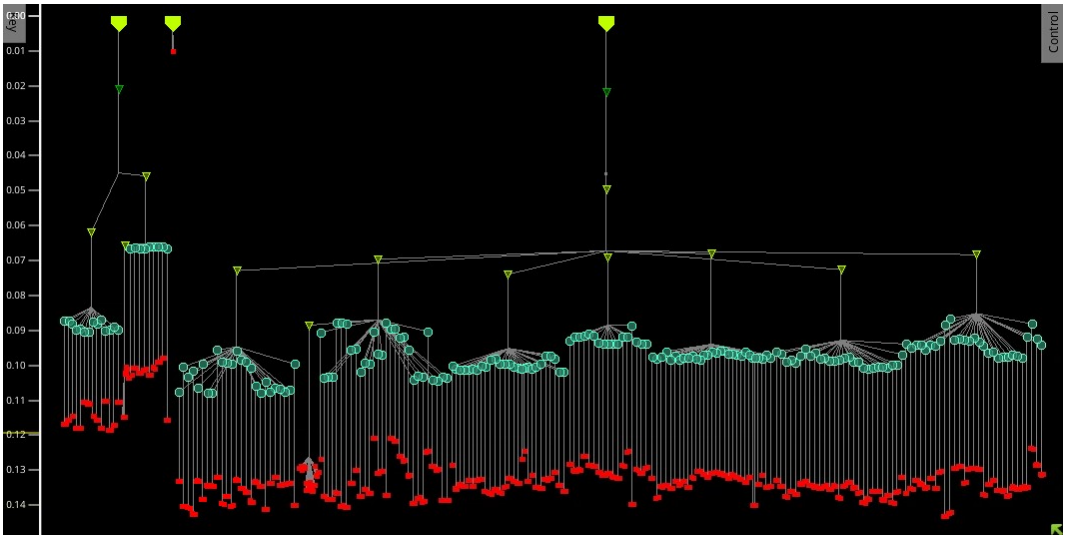
Design Stage	Description	Value
P&R	Number of DRC violations (ex: 0) (Verify -> Verify Geometry…)	0
	Number of LVS violations (ex: 0) (Verify -> Verify Connectivity…)	0
	Die Area (um^2)	3117435.17
	Core Area (um^2)	2265940.09
What type of APR design are you working on? Chip-level design: Netlist with I/O pads Block design: Netlist without I/O pads		Chip-level design

Questions and Discussion

1. Attach the snapshot of scan chain status (command: report_scan_chain) (5%).

```
*** Scan Trace Summary (runtime: cpu: 0:00:00.0 , real: 0:00:00.0):
Successfully traced 3 scan chains (total 1966 scan bits).
Start applying DEF ordered sections ...
Successfully applied all DEF ordered sections.
*** Scan Sanity Check Summary:
*** 3 scan chains passed sanity check.
save out the reporting file "scanChain.rpt"
```

2. Attach the snapshot of CCOpt Clock Tree Debugger result (5%).



3. Attach the snapshot of DRC and LVS checking after routing. (5%)

```
innovus 5> verify_drc
#-check_ndr_spacing auto                # enums={true false auto}, default=auto,  user setting
#-check_same_via_cell true              # bool, default=false, user setting
#-report top.drc.rpt                    # string, default="", user setting
 *** Starting Verify DRC (MEM: 9665.9) ***

  VERIFY DRC ..... Starting Verification
  VERIFY DRC ..... Initializing
  VERIFY DRC ..... Deleting Existing Violations
  VERIFY DRC ..... Creating Sub-Areas
  VERIFY DRC ..... Using new threading
 VERIFY DRC ..... Sub-Area: {226.176 56.544 251.820 84.816} 27 of 81  Thread : 11
 VERIFY DRC ..... Sub-Area: {56.544 56.544 84.816 84.816} 21 of 81  Thread : 10
 VERIFY DRC ..... Sub-Area: {113.088 0.000 141.360 28.272} 5 of 81  Thread : 8
 VERIFY DRC ..... Sub-Area: {56.544 0.000 84.816 28.272} 3 of 81  Thread : 0
 VERIFY DRC ..... Sub-Area: {0.000 56.544 28.272 84.816} 19 of 81  Thread : 2
 VERIFY DRC ..... Sub-Area: {226.176 113.088 251.820 141.360} 45 of 81  Thread : 3
 VERIFY DRC ..... Sub-Area: {169.632 56.544 197.904 84.816} 25 of 81  Thread : 5
 VERIFY DRC ..... Sub-Area: {169.632 0.000 197.904 28.272} 7 of 81  Thread : 4
 VERIFY DRC ..... Sub-Area: {169.632 113.088 197.904 141.360} 43 of 81  Thread : 13
 VERIFY DRC ..... Sub-Area: {226.176 0.000 251.820 28.272} 9 of 81  Thread : 9
 VERIFY DRC ..... Sub-Area: {0.000 113.088 28.272 141.360} 37 of 81  Thread : 1
 VERIFY DRC ..... Sub-Area: {113.088 56.544 141.360 84.816} 23 of 81  Thread : 15
 VERIFY DRC ..... Sub-Area: {56.544 169.632 84.816 197.904} 57 of 81  Thread : 11
 VERIFY DRC ..... Sub-Area: {0.000 169.632 28.272 197.904} 55 of 81  Thread : 14
 VERIFY DRC ..... Sub-Area: {84.816 0.000 113.088 28.272} 4 of 81  Thread : 0
 VERIFY DRC ..... Sub-Area: {0.000 0.000 28.272 28.272} 1 of 81  Thread : 7
 VERIFY DRC ..... Sub-Area: {56.544 113.088 84.816 141.360} 39 of 81  Thread : 12
```

```
VERIFY DRC ..... Sub-Area: {0.000 28.272 28.272 56.544} 10 of 81  Thread : 7
VERIFY DRC ..... Sub-Area: {84.816 141.360 113.088 169.632} 49 of 81  Thread : 10
VERIFY DRC ..... Sub-Area: {113.088 84.816 141.360 113.088} 32 of 81  Thread : 6
VERIFY DRC ..... Sub-Area: {141.360 84.816 169.632 113.088} 33 of 81  Thread : 3
VERIFY DRC ..... Sub-Area: {113.088 141.360 141.360 169.632} 50 of 81  Thread : 10
VERIFY DRC ..... Sub-Area: {84.816 84.816 113.088 113.088} 31 of 81  Thread : 6
VERIFY DRC ..... Sub-Area: {28.272 197.904 56.544 226.176} 65 of 81  Thread : 0
VERIFY DRC ..... Sub-Area: {197.904 197.904 226.176 226.176} 71 of 81  Thread : 14
VERIFY DRC ..... Sub-Area: {0.000 226.176 28.272 251.040} 73 of 81  Thread : 3
VERIFY DRC ..... Thread : 12 finished.
VERIFY DRC ..... Thread : 11 finished.
VERIFY DRC ..... Sub-Area: {141.360 113.088 169.632 141.360} 42 of 81  Thread : 10
VERIFY DRC ..... Thread : 10 finished.
VERIFY DRC ..... Sub-Area: {197.904 28.272 226.176 56.544} 17 of 81  Thread : 9
VERIFY DRC ..... Thread : 9 finished.
VERIFY DRC ..... Sub-Area: {197.904 226.176 226.176 251.040} 80 of 81  Thread : 15
VERIFY DRC ..... Thread : 15 finished.
VERIFY DRC ..... Sub-Area: {197.904 169.632 226.176 197.904} 62 of 81  Thread : 14
VERIFY DRC ..... Thread : 14 finished.
VERIFY DRC ..... Sub-Area: {28.272 169.632 56.544 197.904} 56 of 81  Thread : 0
VERIFY DRC ..... Thread : 0 finished.
VERIFY DRC ..... Sub-Area: {28.272 28.272 56.544 56.544} 11 of 81  Thread : 7
VERIFY DRC ..... Thread : 7 finished.
VERIFY DRC ..... Sub-Area: {28.272 226.176 56.544 251.040} 74 of 81  Thread : 3
VERIFY DRC ..... Thread : 3 finished.
VERIFY DRC ..... Sub-Area: {84.816 113.088 113.088 141.360} 40 of 81  Thread : 6
VERIFY DRC ..... Thread : 6 finished.

 Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:14.0  ELAPSED TIME: 1.00  MEM: 288.1M) ***
```

```
******** Start: VERIFY CONNECTIVITY ********
Start Time: Fri May  1 02:11:56 2026

Design Name: top
Database Units: 2000
Design Boundary: (0.0000, -0.0360) (1765.6200, 1765.5960)
Error Limit = 1000; Warning Limit = 50
Check all nets
```

```
*** 02:11:58 *** Building data for Net VSS
*** 02:11:58 *** Building data for Net VDD
*** 02:11:58 *** Building data for Net VSS
*** 02:11:58 *** Building data for Net VDD
*** 02:11:58 *** Building data for Net VSS

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Fri May  1 02:12:21 2026
Time Elapsed: 0:00:25.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:44.6  MEM: 117.000M)
```

4. Attach the snapshot of the timing report for **setup time and hold time** with no timing violation (post-route). (5%)

```
------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

Setup views included:
 AV_func_0p72v_m40c AV_func_0p72v_125c

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg |reg2cgate| in2reg  | reg2out | in2out  | default |
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.045  |  0.045  |  0.050  |  0.050  |  0.052  |   N/A   |  0.000  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |    0    |   N/A   |    0    |
|          All Paths:|  2116   |  1859   |   207   |   27    |   49    |   N/A   |    0    |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+
```

```
------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

Hold views included:
 AV_func_0p72v_m40c AV_func_0p72v_125c AV_func_0p88v_m40c
 AV_func_0p88v_125c AV_scan_0p88v_m40c AV_scan_0p88v_125c

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg |reg2cgate| in2reg  | reg2out | in2out  | default |
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.001  |  0.001  |  0.016  |  0.012  |  0.083  |   N/A   |  0.000  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |    0    |   N/A   |    0    |
|          All Paths:|  4081   |  3823   |   207   |   28    |   49    |   N/A   |    0    |
+--------------------+---------+---------+---------+---------+---------+---------+---------+
```

5. Show the critical path after post-route optimization. What is the path type? (5%)

   Through command：**report_timing -max_path 1**

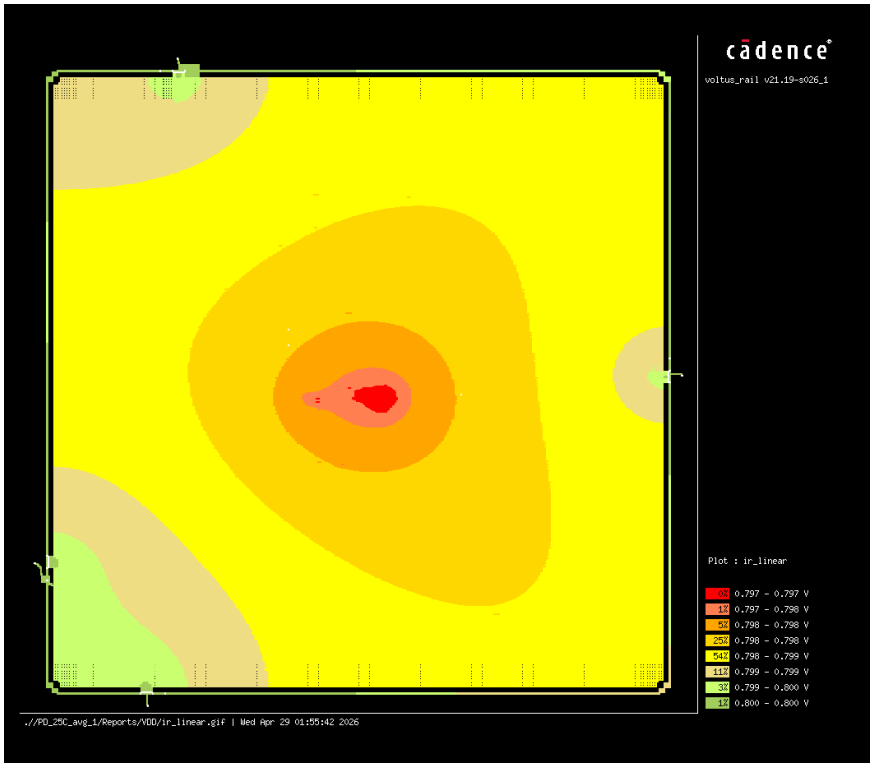
   (The slack of the critical path should match the smallest slack in the timing report)

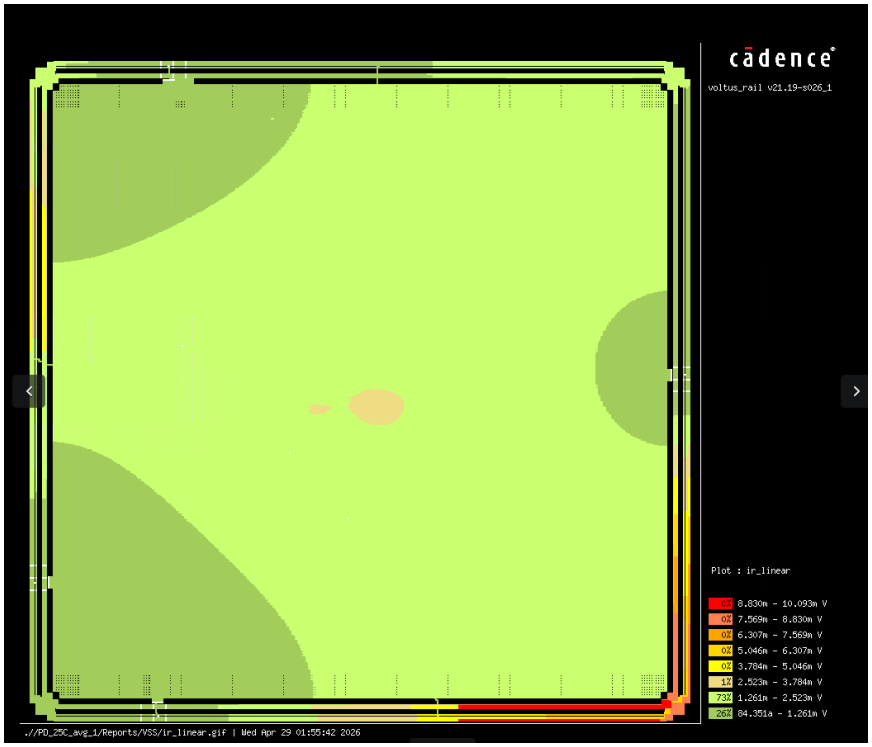
```
Path 1: MET Setup Check with Pin write_ctrl0_axi_write_ctrl0_w_data_buf_reg_3__
1_/CP
Endpoint:   write_ctrl0_axi_write_ctrl0_w_data_buf_reg_3__1_/D (^) checked with
leading edge of 'clk_ram'
Beginpoint: write_ctrl0__cell_315_data_d_int_reg_25_/Q         (^) triggered by
leading edge of 'clk_ram'
Path Groups: {reg2reg}
Analysis View: AV_func_0p72v_125c
Other End Arrival Time          0.143
- Setup                         0.043
+ Phase Shift                   0.500
+ CPPR Adjustment               0.000
- Uncertainty                   0.100
= Required Time                 0.500
- Arrival Time                  0.456
= Slack Time                    0.045
     Clock Rise Edge                 0.000
     + Clock Network Latency (Prop)  0.131
     = Beginpoint Arrival Time       0.131
     +-----------------------------------------------------------------------------------------------------------------+
     |                    Instance                    |     Arc      |       Cell        | Delay | Arrival | Required |
     |                                                |              |                   |       |  Time   |   Time   |
     |------------------------------------------------+--------------+-------------------+-------+---------+----------|
     | write_ctrl0__cell_315_data_d_int_reg_25_       | CP ^         |                   |       |   0.131 |    0.175 |
     | write_ctrl0__cell_315_data_d_int_reg_25_       | CP ^ -> Q ^  | SDFQD1BWP20P90    | 0.070 |   0.201 |    0.245 |
     | FE_PHC8996_write_ctrl0_axi_write_data_25       | I ^ -> Z ^   | BUFFD4BWP20P90LVT | 0.016 |   0.217 |    0.261 |
     | FE_PHC9303_write_ctrl0_axi_write_data_25       | I ^ -> Z ^   | BUFFD4BWP20P90LVT | 0.013 |   0.230 |    0.275 |
     | FE_PHC11113_write_ctrl0_axi_write_data_25      | I ^ -> Z ^   | CKBD1BWP20P90LVT  | 0.014 |   0.244 |    0.289 |
     | FE_PHC10837_write_ctrl0_axi_write_data_25      | I ^ -> Z ^   | CKBD1BWP20P90     | 0.018 |   0.262 |    0.307 |
     | FE_PHC10463_write_ctrl0_axi_write_data_25      | I ^ -> Z ^   | CKBD1BWP20P90     | 0.018 |   0.280 |    0.325 |
     | FE_PHC9932_write_ctrl0_axi_write_data_25       | I ^ -> Z ^   | CKBD1BWP20P90     | 0.031 |   0.311 |    0.356 |
     | FE_PHC8320_write_ctrl0_axi_write_data_25       | I ^ -> Z ^   | CKBD2BWP16P90     | 0.023 |   0.335 |    0.379 |
     | U1336                                          | A1 ^ -> ZN ^ | INR2D1BWP20P90    | 0.020 |   0.354 |    0.399 |
     | FE_PHC8107_write_ctrl0_axi_write_ctrl0_N24     | I ^ -> Z ^   | DEL025D1BWP20P90  | 0.071 |   0.426 |    0.470 |
     | FE_PHC11346_write_ctrl0_axi_write_ctrl0_N24    | I ^ -> Z ^   | CKBD1BWP20P90LVT  | 0.029 |   0.454 |    0.499 |
     | write_ctrl0_axi_write_ctrl0_w_data_buf_reg_3__1_ | D ^        | SDFQD1BWP20P90    | 0.001 |   0.456 |    0.500 |
     +-----------------------------------------------------------------------------------------------------------------+
```

6. Attach the snapshot of GDS stream out messages. (5%)

```
Scanning GDS file /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/GDS/N16ADFP_StdCell.gds to register cell name ......
Scanning GDS file /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdio/N16ADFP_StdIO/GDS/N16ADFP_StdIO.gds to register cell name ......
Scanning GDS file /share1/tech/ADFP/Executable_Package/Collaterals/IP/bondpad/N16ADFP_BondPad/GDS/N16ADFP_BondPad.gds to register cell name ......
Scanning GDS file /share1/tech/ADFP/Lab_adfp/library/memory/rf_2p_hse.gds to register cell name ......
Merging GDS file /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/GDS/N16ADFP_StdCell.gds ......
        ****** Merge file: /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/GDS/N16ADFP_StdCell.gds has version number: 600.
        ****** Merge file: /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/GDS/N16ADFP_StdCell.gds has units: 1000 per micron.
        ****** unit scaling factor = 1 ******
Merging GDS file /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdio/N16ADFP_StdIO/GDS/N16ADFP_StdIO.gds ......
        ****** Merge file: /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdio/N16ADFP_StdIO/GDS/N16ADFP_StdIO.gds has version number: 5.
        ****** Merge file: /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdio/N16ADFP_StdIO/GDS/N16ADFP_StdIO.gds has units: 1000 per micron.
        ****** unit scaling factor = 1 ******
Merging GDS file /share1/tech/ADFP/Executable_Package/Collaterals/IP/bondpad/N16ADFP_BondPad/GDS/N16ADFP_BondPad.gds ......
        ****** Merge file: /share1/tech/ADFP/Executable_Package/Collaterals/IP/bondpad/N16ADFP_BondPad/GDS/N16ADFP_BondPad.gds has version number: 5.
        ****** Merge file: /share1/tech/ADFP/Executable_Package/Collaterals/IP/bondpad/N16ADFP_BondPad/GDS/N16ADFP_BondPad.gds has units: 1000 per micron.
        ****** unit scaling factor = 1 ******
Merging GDS file /share1/tech/ADFP/Lab_adfp/library/memory/rf_2p_hse.gds ......
        ****** Merge file: /share1/tech/ADFP/Lab_adfp/library/memory/rf_2p_hse.gds has version number: 5.
        ****** Merge file: /share1/tech/ADFP/Lab_adfp/library/memory/rf_2p_hse.gds has units: 1000 per micron.
        ****** unit scaling factor = 1 ******
######Streamout is finished!
```

7. Attach the snapshot of IR drop result. (**Make sure IR drop < 8%**) (5%)
   Open **ir_linear.gif** under the path …/PD_25C_avg_x/VDD and VSS. Take the snapshot.





8. Attach the snapshot of the final area result. (5%)

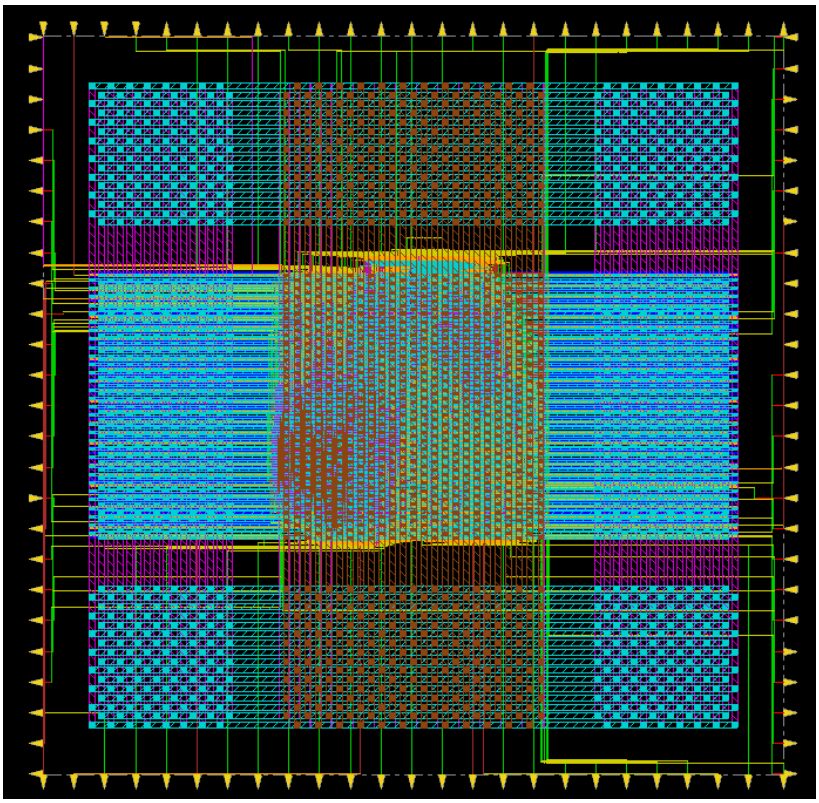
Through command：**analyzeFloorplan**

(analyzeFloorplan destroys the design. Remember to save your design files first!!)

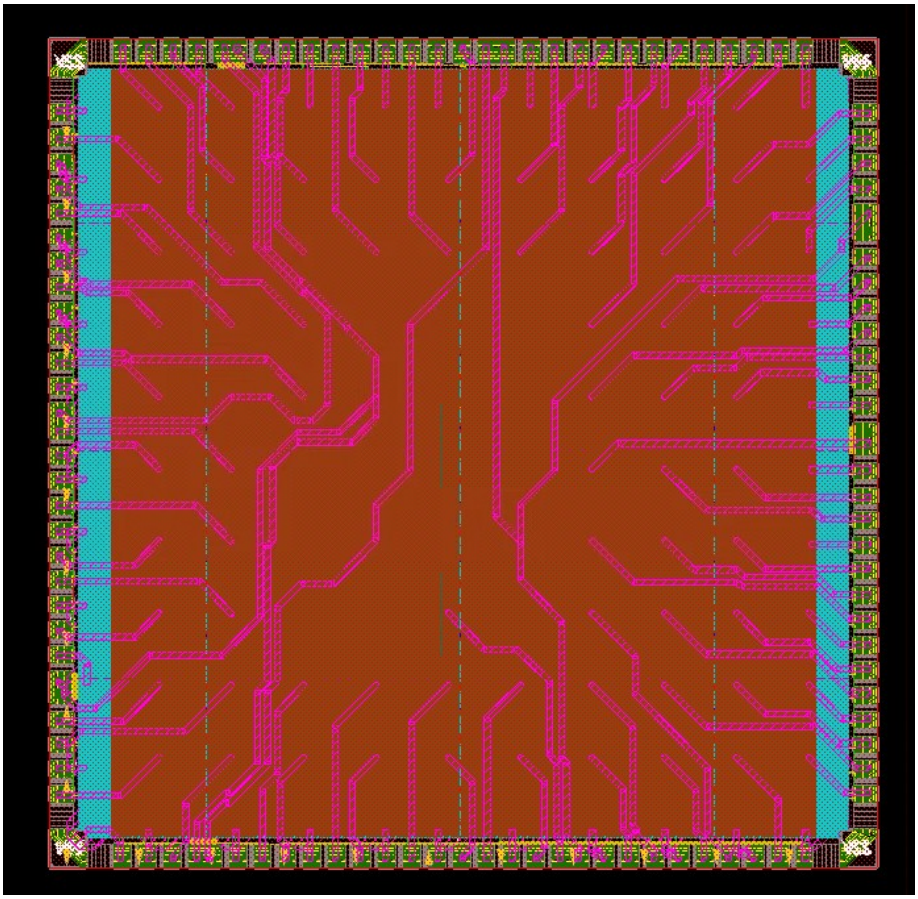
```
************************* Analyze Floorplan **************************
    Die Area(um^2)             : 3117435.17
    Core Area(um^2)            : 2265940.09
    Chip Density (Counting Std Cells and MACROs and IOs): 83.793%
    Core Density (Counting Std Cells and MACROs): 100.000%
    Average utilization        : 100.000%
    Number of instance(s)      : 769427
    Number of Macro(s)         : 0
    Number of IO Pin(s)        : 100
    Number of Power Domain(s)  : 0
************************* Estimation Results *************************
**********************************************************************
```

9. Attach the snapshot of your final layout **after adding core filler**. (Remember to switch to **Physical view** and make Pin Shapes visible) (5%)

Block design example：



Chip-level design example：



10. What is your strategy for bump assignment and for placing signal and power pins/pads? What is the reason behind it? (5%)